Yichao Jin, Ph.D.

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Specialization: Behavioral AI / Health Economics / Behavioral Economics / Discrete Choice Experiments

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Education

University of Texas at Dallas May 2026
Ph.D. in Public Policy and Political Economy

- Successfully Defended: April 2026
- Dissertation: "Intertemporal Choice in Vaccination Timing: Evidence from a Discrete Choice Experiment in Wuhan"
- Committee: Dohyeong Kim (Chair), Richard Scotch, Soyoung Kwon, Khai Chiong.
- Distinction: Dean of Graduate Education Dissertation Research Award (2025).

University of Texas at Dallas May 2026
M.S. in Social Data Analytics and Research

University of Queensland 2019
Master of Development Economics

University of California, Riverside 2017
B.A. in Economics

Research Fields

Primary: Health Economics, Applied Econometrics, Behavioral Economics, Behavioral AI
Secondary: Quantitative Methods, Policy Evaluation, Technology Governance (AI in Health)

Working Papers

1. **"Waiting Time as a Behavioral Barrier to Vaccination Uptake: Nonlinear Heterogeneity by Institutional Trust"**
   Under Review at Economic Analysis and Policy — **Link to Full Draft**
   - Developed a structural Discrete Choice Experiment (DCE) framework to analyze the interaction between temporal barriers and institutional trust in vaccination behavior.
   - Identified nonlinear heterogeneity in waiting time sensitivity, providing a behavioral explanation for vaccine hesitancy.
2. **"Empirical-Frontier Regularization for LLM Synthetic Agents: A Behavioral Digital Twin Framework"**
   Working Paper — **SSRN Link**. Submitted to AI and Economics (AIE) Summer Conference 2026. Developed a 3-stage framework integrating DCE data and Structural Causal Modeling (SCM) to align generative agents with empirical choice frontiers, achieving a 93% reduction in frontier-deviation MSE.

3. **"Value Contamination in Policy Simulation: Measuring How LLM Policy-Value Orientations Shape Predicted Administrative Burden Effects"**
   Working Paper — **SSRN Link**. Develops an audit framework for measuring value sensitivity in LLM-generated policy simulations. Using administrative burden in vaccination access as a testbed, evaluates nine LLMs across six model families through policy-value orientation batteries, narrative frame-decomposition, and within-model frame-manipulation experiments.

4. **"The Price of Waiting: Evidence on Cash–Time Trade-Offs in Vaccination Time Discount"**
   Manuscript in Preparation (Expected submission: August 2026) — Targeting *Social Science & Medicine*.

Research Experience & Affiliations

Doctoral Researcher, Spatial Health AI Research Partnership (SHARP), UT Dallas

- Leading development of Behavioral Digital Twins (BDT): a causally constrained generative framework integrating Discrete Choice Experiments (DCE) with Structural Causal Modeling (SCM) to simulate agent-level behavioral responses to public health policies.
- Managing large-scale datasets and developing computational simulations (R, Python) to evaluate pandemic preparedness.

Graduate Researcher, Behavioral Health Economics Laboratory, UT Dallas

- Conducted high-salience behavioral experiments (N=1,027) analyzing risk-weighted health decisions and institutional trust.

Teaching Experience

Teaching Assistant, School of Economic, Political and Policy Sciences, UT Dallas

- **Quantitative Methods for Policy Analysis (Graduate)**: Led Econometrics labs; specialized in R programming, regression analysis, and causal inference.
- **Health Economics & Public Policy (Undergraduate)**: Developed case studies on health market failures and administrative challenges.
- **Public Policy Analysis (Undergraduate)**: Mentored students in drafting evidence-based policy memos.

Conference Presentations

- **AI and Economics (AIE) Summer Conference 2026**: Empirical-Frontier Regularization for LLM Synthetic Agents.
- **APPAM Fall Research Conference 2025** (Seattle, WA): Estimating Time Discount Rate for Vaccination.
- **DAISY 2026 Workshop at ACM Conference on Health, Inference, and Learning (CHIL) 2026**: Behavioral Digital Twins — A Causally-Faithful Generative Framework.
- **UT Dallas Graduate Research Symposium 2025**: Modeling Vaccination Decisions with Hyperbolic Discounting.

Honors & Awards

- **2025** Dean of Graduate Education Dissertation Research Award, UT Dallas

- **2025** Betty & Gifford Johnson Graduate Travel Award, UT Dallas
- **2016** Omicron Delta Epsilon, International Honor Society for Economics

Technical Skills & Certifications

Econometrics: Mixed Logit (MXL), Latent Class Models (LCM), SEIR Modeling, Causal Inference.
AI & ML: PyTorch, Transformers (HuggingFace), LLM APIs (OpenAI, Anthropic), LangChain, Causal Machine Learning.
Software: Stata (Advanced), R, Python, MATLAB, LaTeX, SQL, GIS.
Certifications: Stanford Artificial Intelligence Graduate Certificate; CITI Human Subjects Protection.

REFERENCES

Dohyeong Kim (Chair)
University of Texas at Dallas
Senior Associate Dean of Graduate Education
Robert E. Holmes Jr. Professor of Public Policy, GIS & Social Data Analytics
Team Lead, SHARP
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